

```

1. public class Arithmeticexc {

    public static void main(String[] args) {

        try {

            int a=10;

            int b=0;

            int result = a / b; // This will throw ArithmeticException

            System.out.println("Result: " + result);

        } catch (ArithmeticException e) {

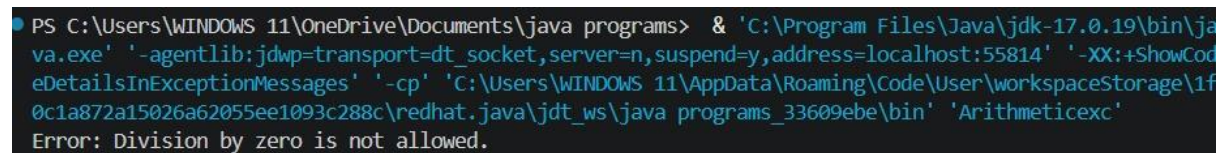
            System.out.println("Error: Division by zero is not allowed.");

        }

    }

}

```



```

PS C:\Users\WINDOWS 11\OneDrive\Documents\java programs> & 'C:\Program Files\Java\jdk-17.0.19\bin\java.exe' '-agentlib:jdwp=transport=dt_socket,server=n,suspend=y,address=localhost:55814' '-XX:+ShowCodeDetailsInExceptionMessages' '-cp' 'C:\Users\WINDOWS 11\AppData\Roaming\Code\User\workspaceStorage\1f0c1a872a15026a62055ee1093c288c\redhat.java\jdt_ws\java programs_33609ebe\bin' 'Arithmeticexc'
Error: Division by zero is not allowed.

```

```

2. public class NullPointerExample {

    public static void main(String[] args) {

        try {

            String str = null;

            System.out.println(str.length());

        } catch (NullPointerException e) {

            System.out.println("NullPointerException;String is null.");

        }

    }

}

```

```

PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> javac NullPointerExample.java
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> java NullPointerExample
NullPointerException;String is null.
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> javac NullPointerExample.java
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> java NullPointerExample
NullPointerException;String is null.
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910>

```

```

3. public class NumberFormatException {
    public static void main(String[] args) {
        try {
            String s = "abc";
            int num = Integer.parseInt(s);
        } catch (NumberFormatException e) {
            System.out.println("Invalid number format.");
        }
    }
}

```

```

PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> javac NumberFormatException.java
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> java NumberFormatException
Invalid number format.
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910>

```

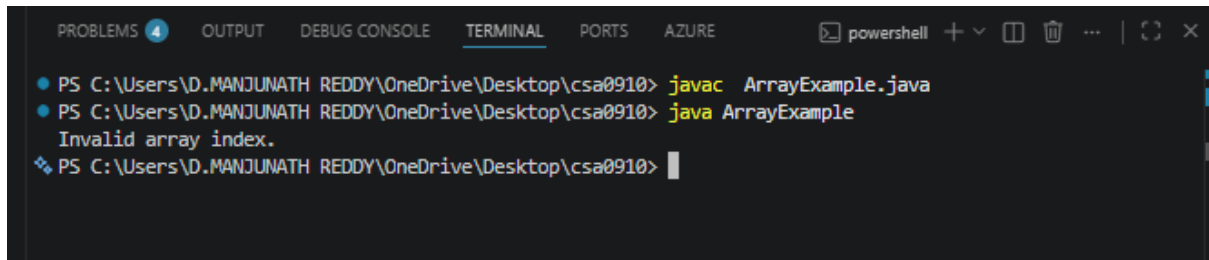
4. ArrayIndexOutOfBoundsException

```

public class ArrayExample {
    public static void main(String[] args) {
        try {
            int[] a = new int[5];
            a[10] = 50; // Invalid index
        } catch (ArrayIndexOutOfBoundsException e) {
            System.out.println("Invalid array index.");
        }
    }
}

```

```
}  
}
```



The screenshot shows a VS Code terminal window with the following commands and output:

```
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> javac ArrayExample.java  
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> java ArrayExample  
Invalid array index.  
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910>
```

5. ArithmeticException – Student Calculator

```
import java.util.Scanner;  
  
public class StudentCalculator {  
    public static void main(String[] args) {  
        Scanner scanner = new Scanner(System.in);  
        System.out.println("=== Student Calculator ===");  
        System.out.print("Enter total marks: ");  
        double totalMarks = scanner.nextDouble();  
        System.out.print("Enter number of subjects: ");  
        int numSubjects = scanner.nextInt();  
        try {  
            // Division by zero will throw ArithmeticException if numSubjects is 0  
            double average = totalMarks / numSubjects;  
            System.out.println("The average marks is: " + average);  
        } catch (ArithmeticException e) {  
            System.out.println("Error: Number of subjects cannot be zero. Please enter a valid number.");  
        } finally {  
            scanner.close();  
        }  
    }  
}
```

```

• PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> javac StudentCalculator.java
• PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> java StudentCalculator
=== Student Calculator ===
Enter total marks: 300
Enter number of subjects: 0
The average marks is: Infinity
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> 

```

6. NullPointerException

```

class Employee {
    int empld;
    String name;

    public Employee(int empld, String name) {
        this.empld = empld;
        this.name = name;
    }

    public void displayDetails() {
        System.out.println("Employee ID: " + empld);
        System.out.println("Employee Name: " + name);
    }
}

public class EmployeeManagementSystem {
    public static void main(String[] args) {
        System.out.println("=== Employee Management System ===");

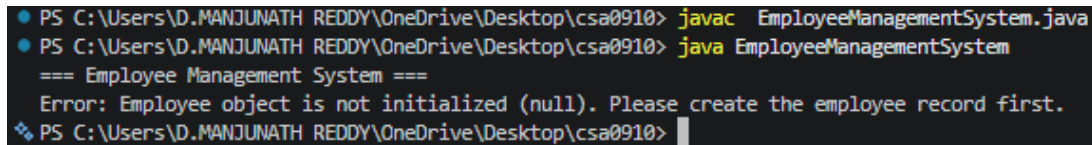
        // Declaring an Employee reference but intentionally leaving it uninitialized (null)
        Employee emp = null;

        try {
            // Attempting to access a method on a null object causes NullPointerException
            emp.displayDetails();
        } catch (NullPointerException e) {

```

```
        System.out.println("Error: Employee object is not initialized (null). Please create the employee record first.");
```

```
    }  
}  
}
```



```
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> javac EmployeeManagementSystem.java  
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> java EmployeeManagementSystem  
=== Employee Management System ===  
Error: Employee object is not initialized (null). Please create the employee record first.  
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910>
```

```
import java.util.Scanner;
```

```
public class StudentMarksProcessing {
```

```
    public static void main(String[] args) {
```

```
        // Storing marks of 5 students in an array (valid indices: 0 to 4)
```

```
        int[] marks = {85, 90, 78, 92, 88};
```

```
        Scanner scanner = new Scanner(System.in);
```

```
        System.out.println("=== Student Marks Processing ===");
```

```
        System.out.println("Stored marks for 5 students.");
```

```
        System.out.print("Enter the index of the student (Valid range: 0 to 4): ");
```

```
        int index = scanner.nextInt();
```

```
        try {
```

```
            // Accessing array element outside bounds throws ArrayIndexOutOfBoundsException
```

```
            System.out.println("Mark at index " + index + " is: " + marks[index]);
```

```
        } catch (ArrayIndexOutOfBoundsException e) {
```

```
            System.out.println("Error: Entered index is outside the valid range.");
```

```
            System.out.println("Please enter an index between 0 and " + (marks.length - 1) + ".");
```

```
        } finally {
```

```
            scanner.close();
```

```

    }
}
}

```

```

• PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> javac StudentMarks.java
• PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> java StudentMarks
=== Student Marks Processing ===
Stored marks for 5 students.
Enter the index of the student (Valid range: 0 to 4): 6
Error: Entered index is outside the valid range.
Please enter an index between 0 and 4.
❖ PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910>

```

8. NumberFormatException

```

import java.util.Scanner;

public class CustomerRegistrationSystem {

    public static void main(String[] args) {

        Scanner scanner = new Scanner(System.in);

        System.out.println("=== Customer Registration System ===");

        System.out.print("Enter customer's age as a string: ");

        String ageString = scanner.next();

        try {

            // Converting String to integer; throws NumberFormatException if input is non-numeric
            int age = Integer.parseInt(ageString);

            System.out.println("Customer registration successful! Age recorded: " + age);

        } catch (NumberFormatException e) {

            System.out.println("Error: Invalid input format entered for age (" + ageString + ").");

            System.out.println("Please enter numeric digits only (e.g., 25).");

        } finally {

            scanner.close();

        }

    }

}

```

```
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> javac CustomerRegistrationSystem.java
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> java CustomerRegistrationSystem
=== Customer Registration System ===
Enter customer's age as a string: AB
Error: Invalid input format entered for age ('AB').
Please enter numeric digits only (e.g., 25).
PS C:\Users\D.MANJUNATH REDDY\OneDrive\Desktop\csa0910> |
```